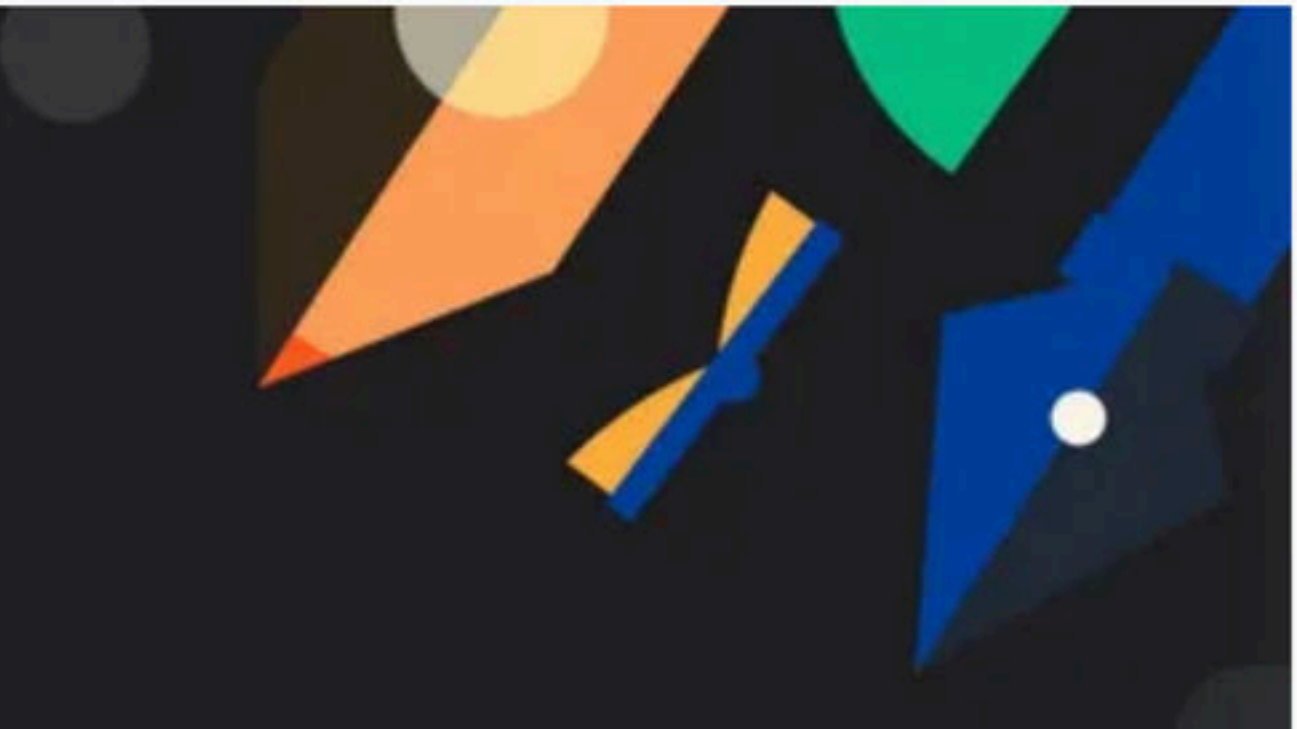
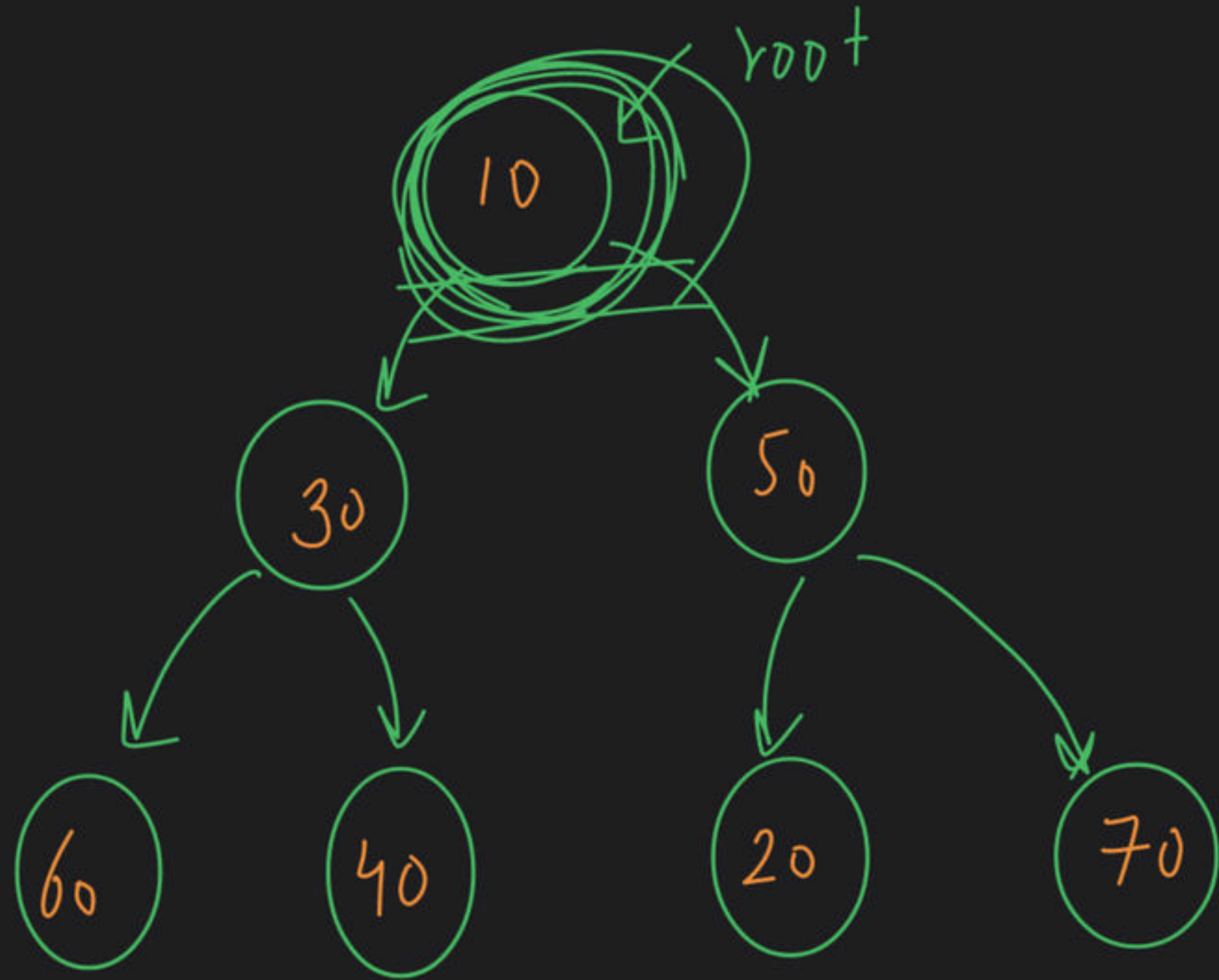




Trees Class - 3

Special class

→ Construct a tree from inorder & preorder traversal



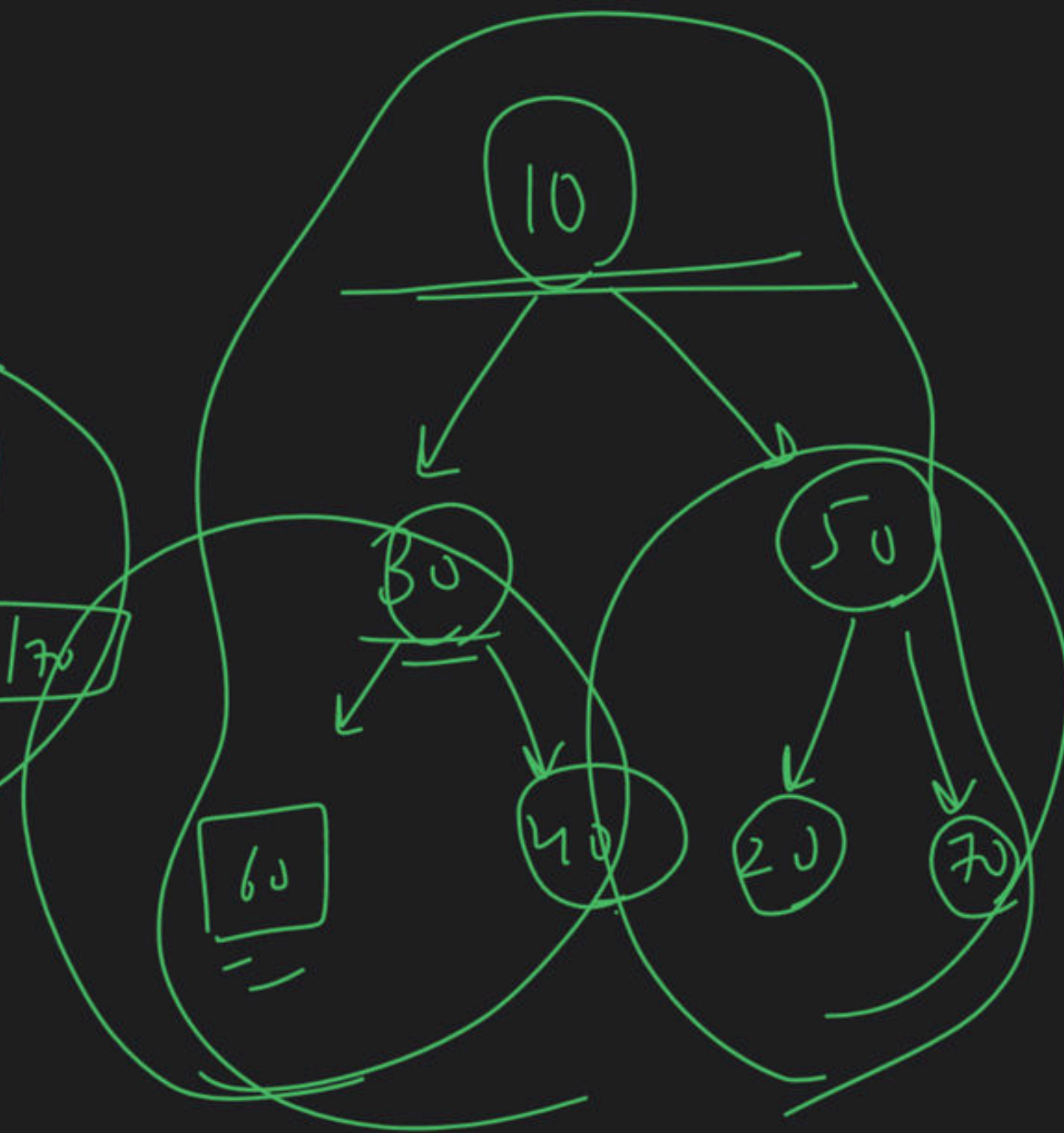
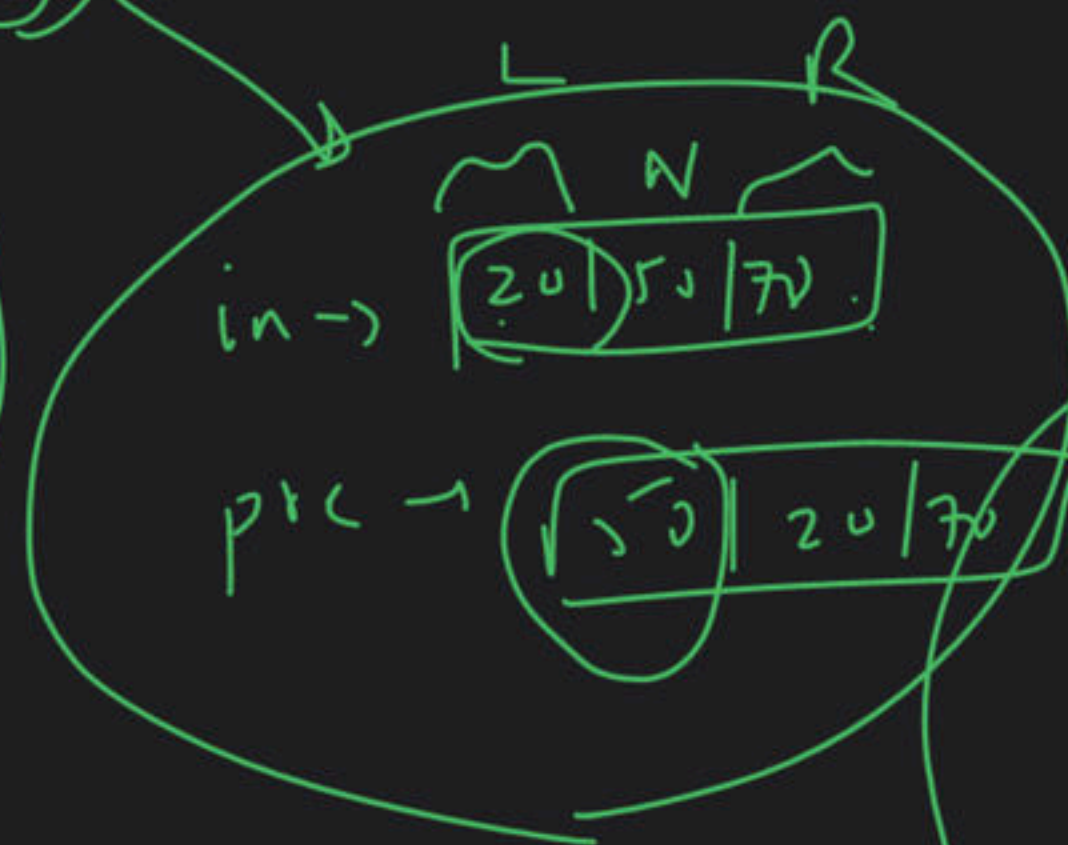
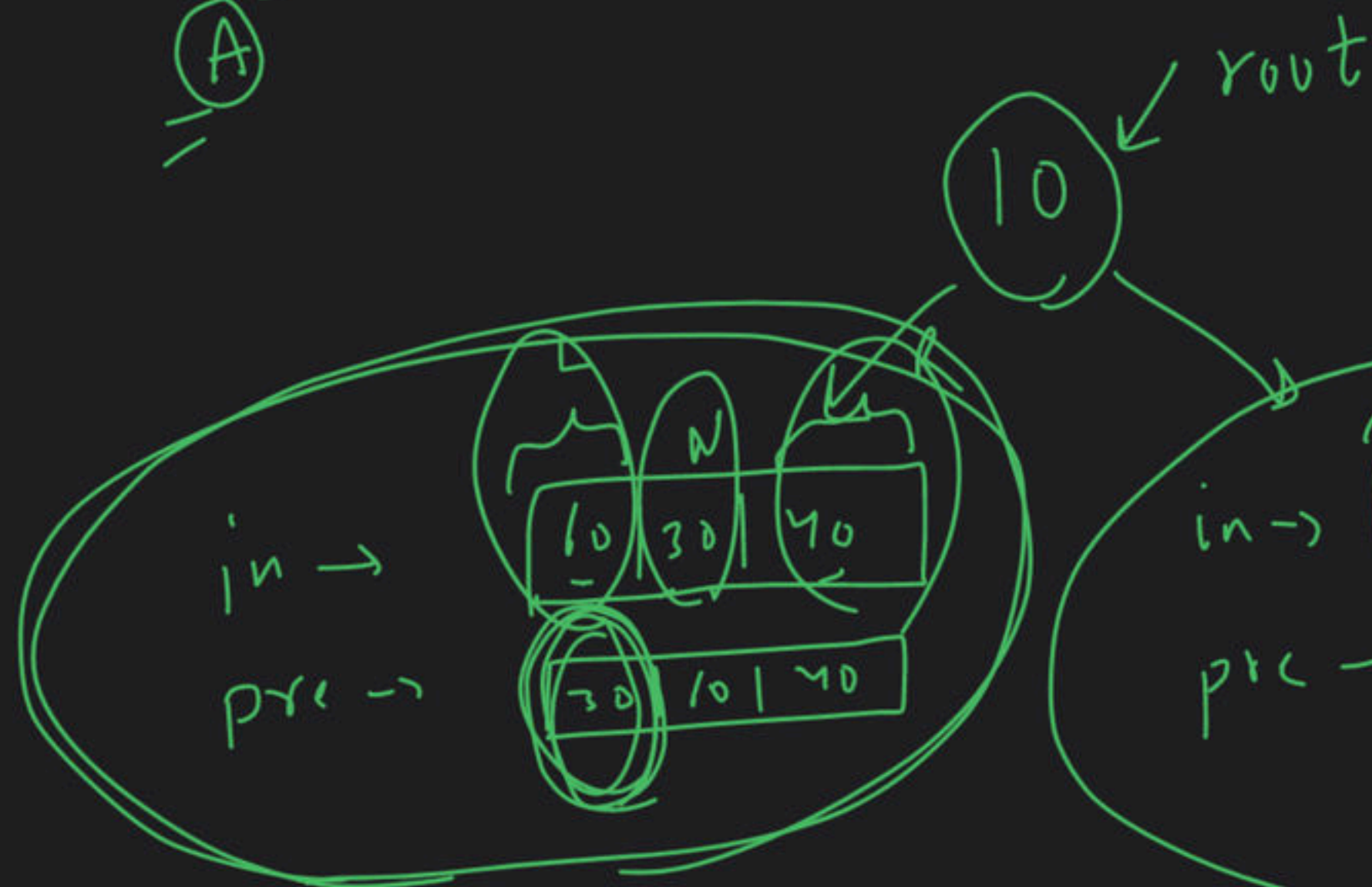
(LNR)
inorder → 60 30 40 10 20 50 70

(NLNR)
preorder → 10 30 60 40 50 20 70
root

= inorder $\rightarrow [10, 30, 40, 10, 20, 50, 70] \rightarrow [L \ N \ R]$

= preorder $\rightarrow [10, 30, 60, 40, 50, 20, 70] \rightarrow [N \ L \ R]$

A



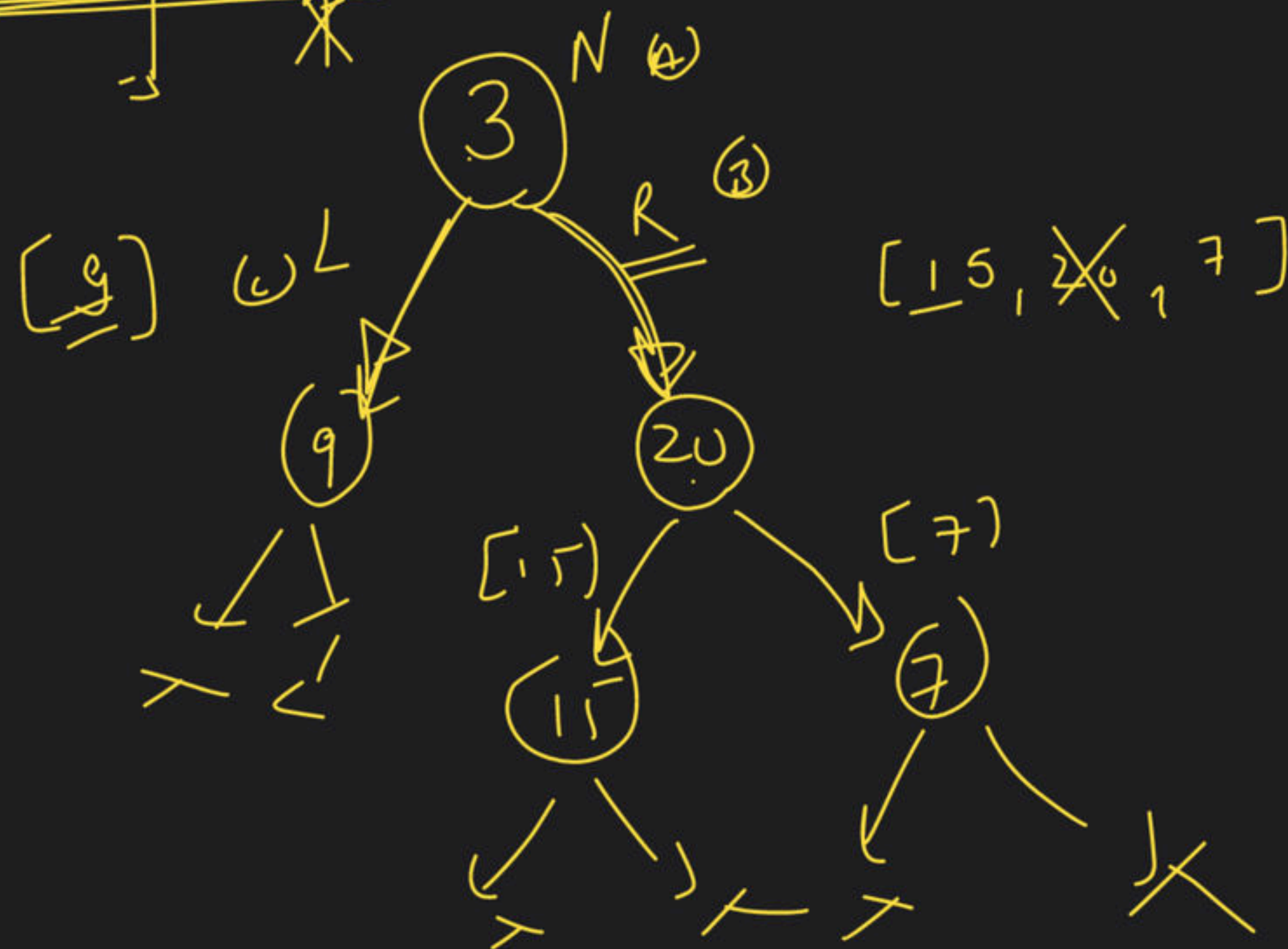
$i_n \rightarrow$

in $\rightarrow$ $\overset{L}{\boxed{9}} \cancel{3}, \overset{R}{\boxed{15, 20, 7}} \rightarrow \underline{L} \underline{N} R$

post = $\boxed{9}, \boxed{15}, \boxed{7}, \boxed{20}, \boxed{3} \rightarrow \underline{L} \underline{R} \underline{N}$

in/post

right subtree
left subtree

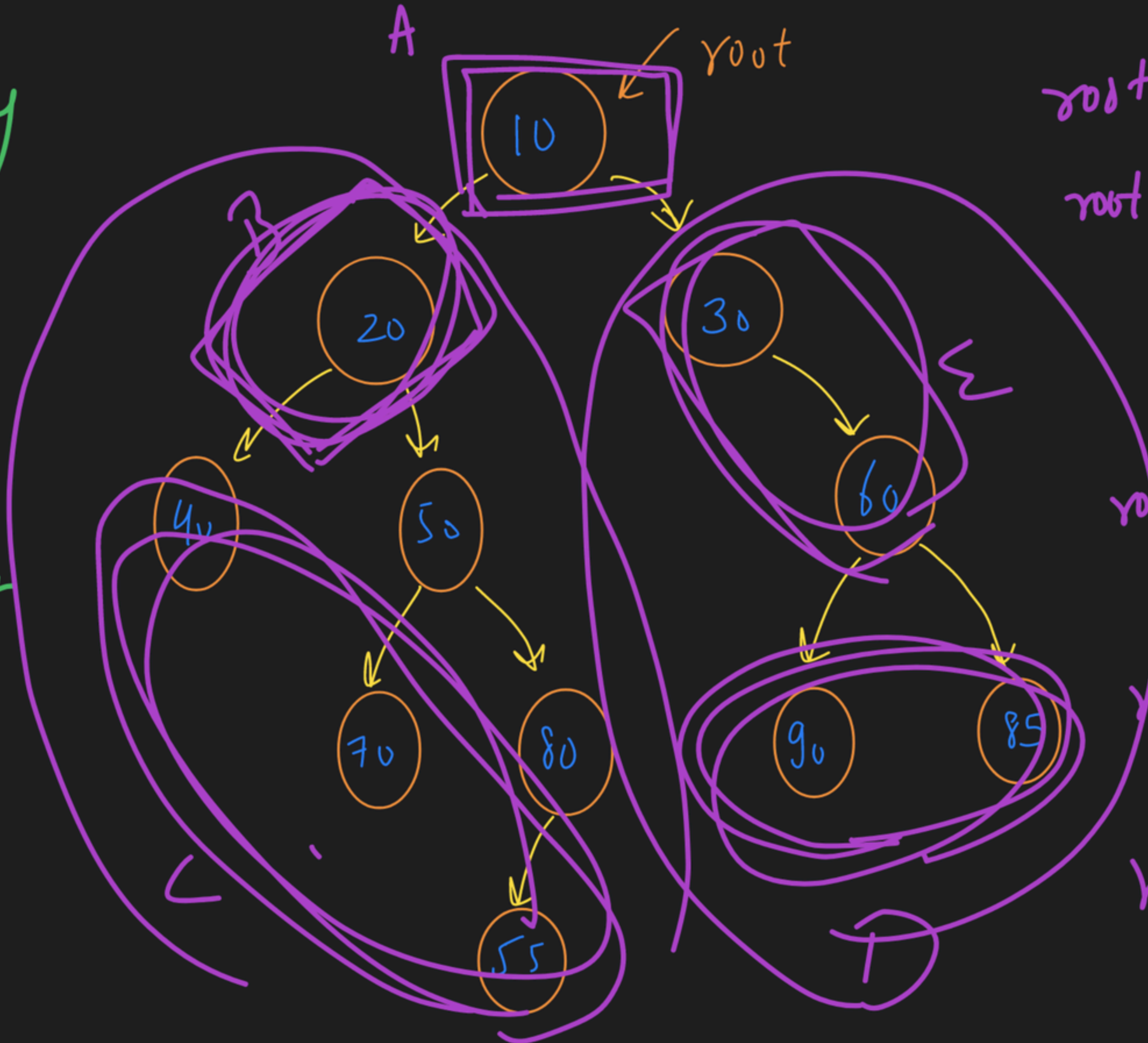


Boundary

→ Left B

→ Leaf B

→ Rev Right



root Node

root → left

→ left B

→ Leaf A

root → left

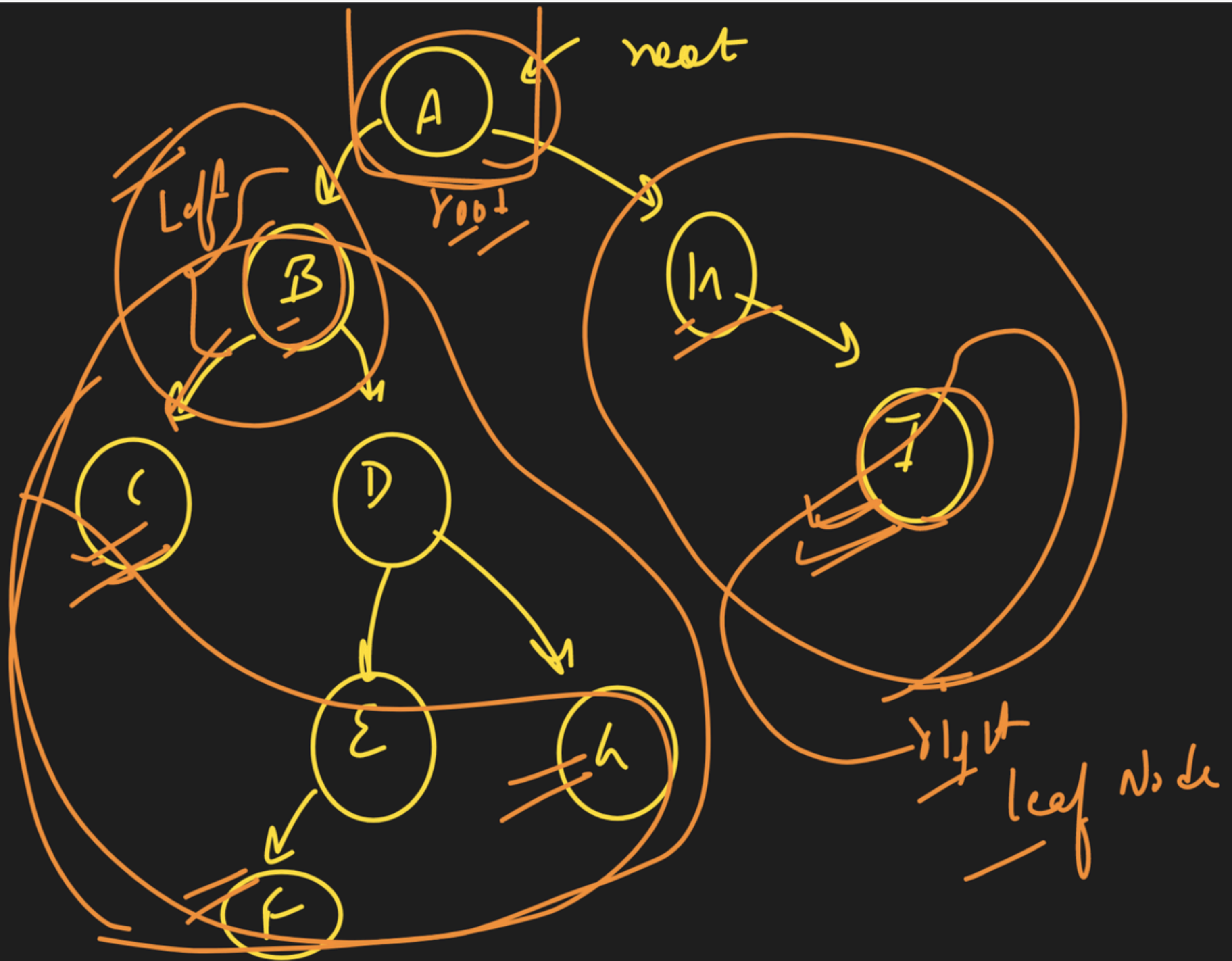
→ Leaf B

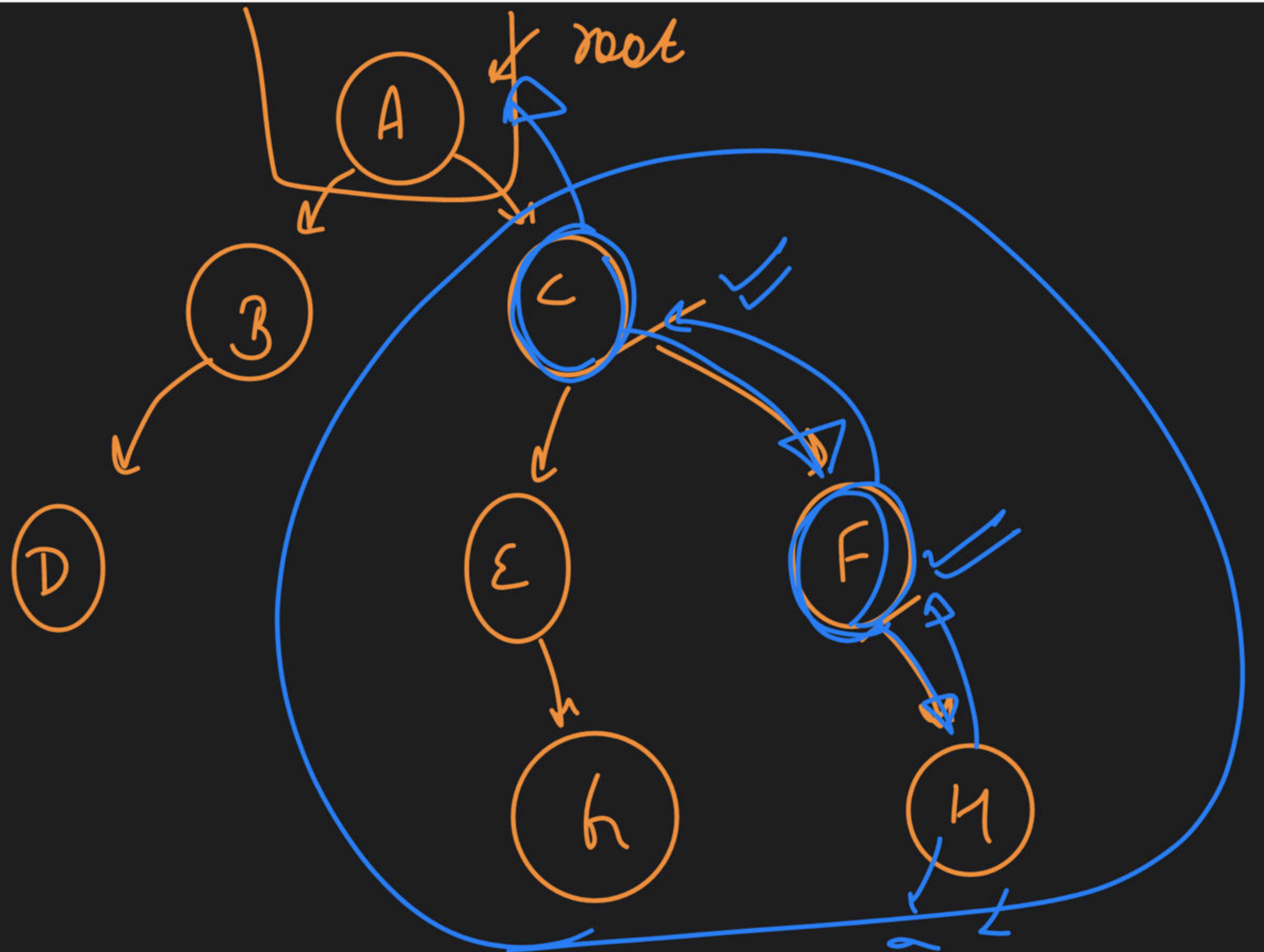
root → right

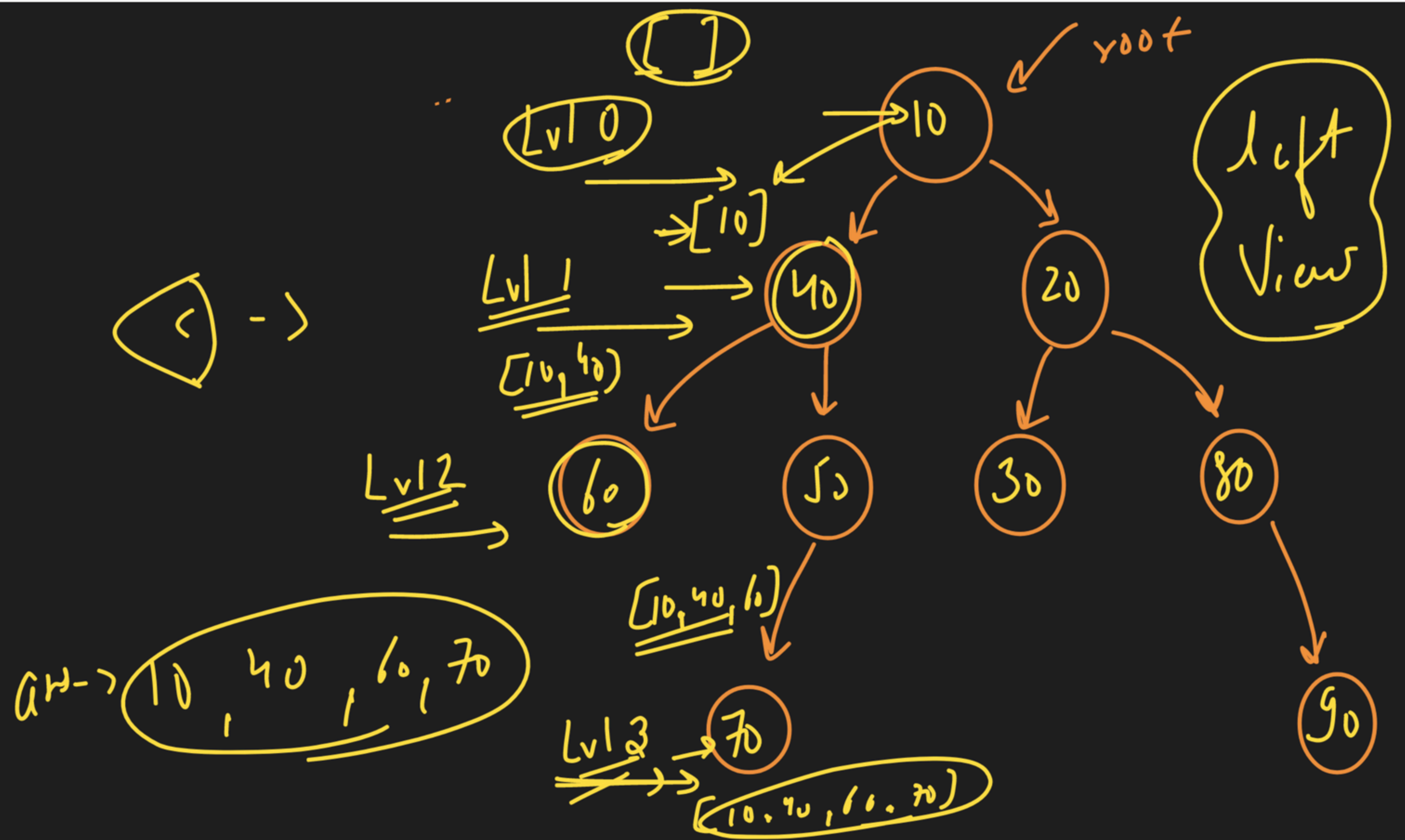
→ Leaf B

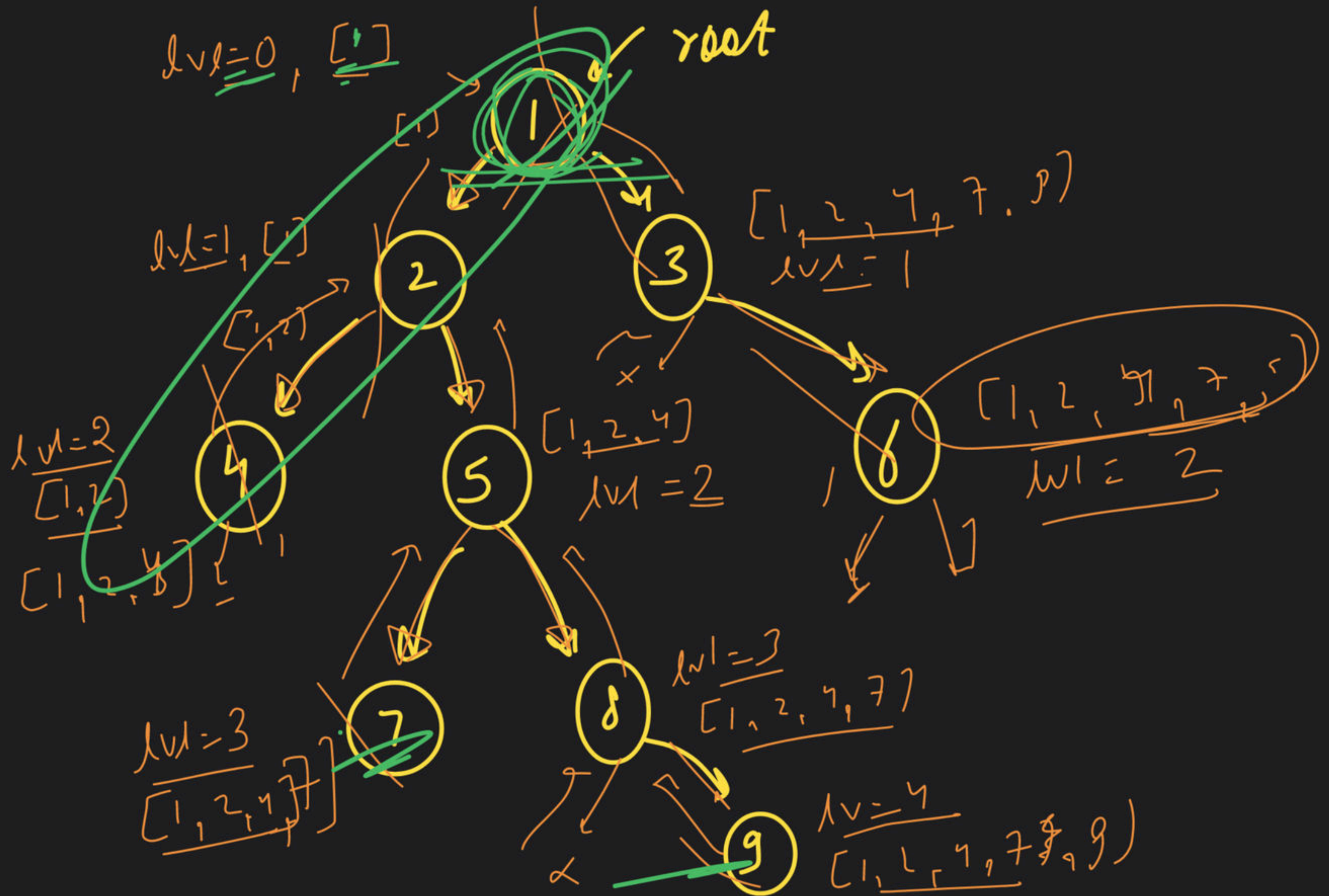
root → right

Left
leaf node









In α
 pre α
 post α

